A FACE AT THE SCIENCE-LAB DOOR

Building, deploying and judging a Keras image model for an urban junior school — and why the cloud must stay small around children

1. The school's Tuesday problem

Riverside Junior sits on a busy inner-city junction. Two thousand children aged seven to eleven pour through three gates between 08:20 and 08:45. Mrs Adeyemi in the front office knows hundreds of faces. She does not know two thousand. Last term a Year 4 boy, late and eager to help, followed a delivery driver through the staff car-park gate and was found standing in the chemical prep room because the inner door had been propped open with a fire extinguisher. Nobody shouted. The science technician simply went grey.

The head asked for "something on the door that knows our children." Restricted rooms are real: the prep room, the pool changing corridor, the server cupboard, the roof-plant stairs. Supply teachers change weekly. A lanyard can be borrowed. A face, the argument went, cannot.

That sentence is where the work of this paper begins — and where it has to slow down. Identifying children with a camera is not the same as classifying a picture of a truck. It is biometric processing of minors. Andrejevic and Selwyn (2020) warn that facial recognition in schools can harden a place of learning into a place of sorting. The technical brief was therefore rewritten before a single layer was stacked: prove that an open-source Convolutional Neural Network (CNN) can be built, trained, exported and hosted on a cloud pattern the school could afford; do it without a photograph of any child; and be honest about what the numbers do and do not mean.

2. What was actually built

2.1 Tools and constraint

The stack the brief named — TensorFlow and Keras — is the right teaching stack: both are open source, both speak the same Sequential API, both export weights a cloud job can reload. In this lab TensorFlow's wheel would not install, so Keras 3.15.1 ran on a PyTorch backend. The Python is the same code a TensorFlow runtime would accept. No image of a pupil was downloaded, scraped or generated. Training used 4,000 synthetic 32×32 colour tiles: each of ten fictional "zones" (science lab, pool changing, library door, and so on) received a colour field and a stripe. The model's job was to name the zone, not the child. That is a pipeline rehearsal, not a production identity system.

2.2 Model

A three-block convolutional network was compiled with Adam and sparse categorical cross-entropy:

- Input 32×32×3, rescaled to [0, 1].
- Conv2D 32 → MaxPool; Conv2D 64 → MaxPool; Conv2D 128.
- Global average pooling, dropout 0.35, dense 64, softmax over 10 zones.
- 102,154 trainable parameters — small enough for a CPU evening and for an edge device later.

Training ran four epochs, batch 64, 15 per cent validation split. Wall time was 12.3 seconds. Training accuracy climbed 0.44 → 0.87 → 0.98 → 0.997. Validation accuracy reached 1.00 by epoch three. On a held-out 800-tile test set, accuracy was 1.00 and loss 0.0036. Every zone recall was 1.00. Those figures are expected on an artificially

clean task. They prove the loop — load, compile, fit, evaluate, save weights — works. They do not prove a camera can tell Amina in Year 5 from her twin in Year 4 under a wet hood at 08:31.

2.3 How a live school system would differ

A production design would not classify "who is this child?" from a softmax over 2,000 names. It would detect a face, compute a compact embedding (MobileNetV2 or a pruned ResNet, as transfer-learning studies using Keras routinely do — Anwarul, Choudhury and Dahiya, 2023), and compare that vector to a tightly scoped gallery of pupils who have explicit, renewable parental consent for a named door. A threshold plus a human override sits on the lock. Images are not stored after the embedding is made. The gallery lives in a regional cloud vault with customer-managed keys. False rejects must fail closed on the prep-room door and fail open on a fire-exit. Banzon, Beever and Taub (2024) ask schools to treat facial pipelines as an ethical practice, not only an accuracy race; that is the procedure this design follows.

Stage	Lab (this paper)	School cloud (proposed)	Why the gap exists
Data	4,000 synthetic tiles	Consented embeddings only	Children's faces are special-category data
Model	3-block CNN, 102k params	MobileNetV2 + embedding head	Need lighting / pose robustness
Host	Local Keras 3 / torch	Edge infer + regional GPU burst	08:20 spike; idle by 09:15
Decision	Softmax zone label	Cosine match + steward override	A lock is not a leaderboard

3. Cloud deployment pattern

The school does not need a permanent GPU. It needs a quiet night-time training job when a new Year 3 cohort arrives, and a cheap inference path at the three gates between 08:15 and 09:00. The pattern:

- Train: a preemptible GPU notebook (Vertex AI Workbench, SageMaker Studio, or Azure ML) scheduled in the evening. Checkpoint weights to object storage. Delete the training machine when the job ends.
- Package: export SavedModel / ONNX. Quantise to INT8 so a Raspberry-class edge box at the door can run the embedding without sending raw frames off-site.
- Serve: Cloud Run or Cloud Functions for the rare "who is at the door?" call that the edge box cannot settle. Autoscale to zero after the morning peak.
- Govern: customer-managed keys, regional residency, audit log of every unlock, 12-month deletion of embeddings when a child leaves, Data Protection Impact Assessment on the file before go-live.

That is the shared-responsibility model in school language. The vendor keeps the hall lights on. The school decides whose face is allowed to turn them on.

4. Evaluation — what we measured, what we refused to claim

We evaluated the lab model the way a teaching exercise should: loss and accuracy on a held-out set, per-class recall, parameter count, and wall-clock time. The network learned the synthetic zones completely. That is a successful unit test of the Keras loop. It is a failed test of the original wish ("identify 2,000 children"). Real faces

vary by rain, hats, siblings, and growth spurts. Classroom face systems in the literature only look strong on their own datasets; they sag when the lighting changes. A school gate is that change, every morning.

So evaluation for Riverside, if the governors ever approved a consented embedding system, would add: false-accept rate at the prep-room door (must be vanishing); false-reject rate at the library (annoying but survivable); latency under 300 ms; a steward's override log; and a termly bias check across year groups and skin tones. Accuracy without those numbers is a brochure.

5. Reflection: building, deploying, judging

Building the model was the easy hour. Keras hid the gradients. Three convolutional blocks, a dropout line, Adam. The first two training runs lied to me: batch-normalisation on an overly simple tile set drove validation accuracy down to chance while training accuracy sat at 1.00. That is a useful bruise. A school that buys a "99 per cent accurate" camera without watching the validation curve is buying a story. I stripped the normalisation, retrained, and the curve became honest. Deployment thinking changed the architecture more than accuracy did. Once I pictured Mrs Adeyemi at 08:22 with a queue in the rain, I stopped wanting a fat softmax over two thousand names and started wanting a small embedding that could live on a box above the door. Evaluation then stopped being a leaderboard and became a risk conversation: which error is worse, locking a child out of the library or letting a stranger into the chemicals? Those are not hyperparameters. They are values.

I also had to refuse part of the brief. A working recogniser of real pupils would have required a gallery of children's faces. I do not have lawful, proportionate grounds to create one in a coursework lab, and the literature is not shy about the harm of pretending otherwise (Andrejevic & Selwyn, 2020). The ethical constraint became a design constraint: synthetic zones in the lab, embeddings-plus-consent in any future live system, human override on every lock.

6. Reflection: how AI changes cloud operations

The same morning spike that frightens Mrs Adeyemi is the thing cloud operations already know how to name: a bursty, diurnal workload. Reactive autoscaling waits for CPU to trip a threshold and then arrives late. Predictive, machine-learning autoscalers choose the metrics that actually move the application and provision ahead of the surge. Pintye, Kovács and Lovas (2024) show that selecting the right signals — not merely watching average CPU — cut quality-of-service violations by up to 80 per cent and virtual-machine use by 3–50 per cent. For Riverside that translates into a concrete bill: GPU minutes at 07:50 to warm the embedding service, scale-to-zero by 09:15, a cheap CPU path for the rest of the day. Training stays on preemptible machines. Inference stays at the edge so raw frames of children never ride the public internet.

AI helps the cloud in three school-sized ways. First, it forecasts the gate. A small time-series model on last term's arrival counts is enough to pre-warm three containers rather than thirty. Second, it shrinks the model itself — pruning and quantisation — so the school pays for a four-watt box instead of a rented T4 all day. Third, it watches its own cost: if false rejects climb on wet mornings, the operations loop scales a heavier model for an hour instead of running the heavy model forever. Resource management stops being a guess by the ICT technician on a Sunday night and becomes a closed loop: measure the door, predict the crowd, place the compute where the children actually are, and turn it off when they sit down for assembly.

That is the modest version of “AI on the cloud” this school can defend to a parent. Not a camera that owns a child’s face. A door that is locked when it should be, a bill that sleeps after registration, and a record of who decided what.

References

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Lab artefacts: school_access_cnn.py and model_lab/riverside_zone_cnn.weights.h5 (Keras 3.15.1, 102,154 parameters, synthetic zone tiles only — no images of children).